5G MAG The Media Connectivity Association




OVERVIEW AND UPDATES · REFERENCE TOOLS

# 3GPP RAN and Core Platforms

Lab-ready 5G RAN and Core built on Open-Source platforms

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# About the project

The specifications behind the 5G System, the radio access network and the services the libraries consume

## 3GPP TS 23.501

**Technical Specification, V20.3.0**

System architecture for the 5G System (5GS), Stage 2: the Network Functions and reference points of the 5G Core.

## 3GPP TS 38.300

**Technical Specification, V19.4.0**

NR and NG-RAN Overall Description, Stage 2.

## 3GPP TS 38.401

**Technical Specification, V19.3.0**

NG-RAN architecture description.

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More information at **5g-mag.com/tech/5g-core/overview**

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# What the platforms provide

The network infrastructure that the other Reference Tools run against

## NG-RAN

The gNB, split into CU and DU, built from rt-srsRAN_Project, the fork of the srsRAN Project.

## 5G Core

Built from 5G-MAG's fork of Open5GS. The multicast and broadcast core functions belong to the MBS project.

## LTE and UE

Built from srsRAN_4G: the basis for the LTE-based 5G Broadcast components and for UE functions for MBS.

## SDR

Software Defined Radio front-ends transmit and receive the radio signal that the software generates and processes.

**2** **delivery technologies**: MBS (5G NR with the 5G Core) and LTE-based 5G Broadcast. The forks follow this split. The description of the forks is not yet audited.

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# Two delivery technologies

Why the forks split

MBS

## 5G NR and the 5G Core

Multicast and broadcast functions deliver the same content to many devices. The RAN software basis is rt-srsRAN_Project (NR CU/DU).

5G BROADCAST (LTE-BASED)

## LTE/eMBMS radio

Reuses FeMBMS, with no core and no return channel, for free-to-air TV and radio, receive-only. The RAN basis is srsRAN_4G (eNodeB with MBMS).

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## Open source bases

The srsRAN Project and srsRAN_4G of Software Radio Systems for the radio, and the Open5GS project for the 5G Core.

**The dedicated 5G Broadcast transmitter and modem are in separate repositories (rt-mbms-tx and rt-mbms-modem), which depend on this project for the radio software.**

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# 5G Core Service Consumers

Reusable libraries that let an application call 5G Core services

BSF

## libscbsf

Finds the BSF through the NRF, then which PCF manages the PDU Session of a UE, identified by its IP address (Nbsf_Management).

PCF

## libscpcf

Lets an application request an application session context from the PCF, to change the network QoS of selected traffic flows of a PDU Session (Npcf_PolicyAuthorization, at N5). Tool: pcf-policyauthorization.

MB-SMF

## libscmbsmf

Allocates and deallocates TMGIs (Nmbsmf_TMGI), and creates, modifies and destroys MBS Sessions with their event subscriptions (Nmbsmf_MBSSession). Tools: tmgi-tool and mbs-service-tool.

**Each library implements only the consumer side of its APIs, so it needs a running 5G Core. 5G Media Streaming uses the BSF and PCF libraries; MBS uses the MB-SMF library.**

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# Services for 5G Media Streaming

The service operations of the PCF, the BSF and the NRF that the BSF and PCF libraries use

**PCF** TS 29.514 V20.1.0, clause 4.2: Npcf_PolicyAuthorization

Create | Update | Delete | Notify | Subscribe | Unsubscribe

**BSF** TS 29.521 V20.1.0, clause 4.2: Nbsf_Management

Register | Deregister | Discovery | Update | Subscribe | Unsubscribe | Notify

**NRF** TS 29.510 V20.1.0, clause 5.3.2: Nnrf_NFDiscovery

NFDiscover | SCPDomainRoutingInfoGet | SCPDomainRoutingInfoSubscribe | SCPDomainRoutingInfoNotify | SCPDomainRoutingInfoUnSubscribe

The specifications define these operations. Only the Network Functions that 5G Media Streaming and 5G MBS use are covered.

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# Services for 5G MBS

The service operations of the MB-SMF that the MB-SMF library uses

**Nmbsmf_TMGI** TS 29.532 V20.0.0, clause 5.2.2

- TMGI Allocate
- TMGI Deallocate

**Nmbsmf_MBSSession** TS 29.532 V20.0.0, clause 5.3.2, at reference point Nmb1

- Create
- Update
- Release
- ContextUpdate
- StatusSubscribe
- StatusUnsubscribe
- StatusNotify
- ContextStatusSubscribe
- ContextStatusUnSubscribe
- ContextStatusNotify

The library finds the MB-SMF through the NRF discovery service (Nnrf_NFDiscovery).

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# What later releases changed

What each 3GPP release added to the services the libraries consume

| Release 16 | Release 17 | Release 18 | Release 19 |
| --- | --- | --- | --- |
| PCF: QoS monitoring for service data flows. BSF: several UE addresses in one binding, and binding update. | MB-SMF: TS 29.532 approved at version 17.0.0. NRF: SCP domain routing information, and MB-SMF registration and discovery. BSF: subscription to PCF registration, and discovery of a PCF for a UE. | MB-SMF: Associated Session ID, RedCap UE information and MOCN support. PCF: multi-modal, PDU set and L4S support for XR and media services. NRF: shared data. | PCF: the MPS for Messaging indication on N5. NRF: selective parameter omission in responses. BSF: PCF as a consumer of discovery, and BSF subscription expiry time. |

Release 20 (work plan of 28 September 2026): work items list TS 29.510 and TS 29.514; none lists TS 29.521 or TS 29.532.

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# Repositories

The repositories of the 3GPP RAN and Core Platforms

| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| rt-5gc-service-consumers | reusable service consumer libraries for the 5GC Network Functions, with command line tools | BSF, PCF and MB-SMF libraries | ● 2.1.1 5 Jun 2026 | GitHub |
| rt-srsRAN_Project | fork of the srsRAN Project, for the development and adaptation of NG-RAN procedures | NG-RAN (gNB) | ○ none yet | GitHub |
| srsRAN_4G | fork of srsRAN_4G, for the development and adaptation of NG-RAN procedures | LTE radio and UE | ○ none yet | GitHub |
| srsRAN | fork of the discontinued srsRAN suite, kept for reference; superseded by srsRAN_4G and rt-srsRAN_Project | Kept for reference | ○ none yet | GitHub |
| open5gs | fork of Open5GS, for the development and adaptation of 5GC network functions | 5G Core | ● v0.1.4-5mbs-v2.7.2-o5gs 14 Jul 2026 | GitHub |

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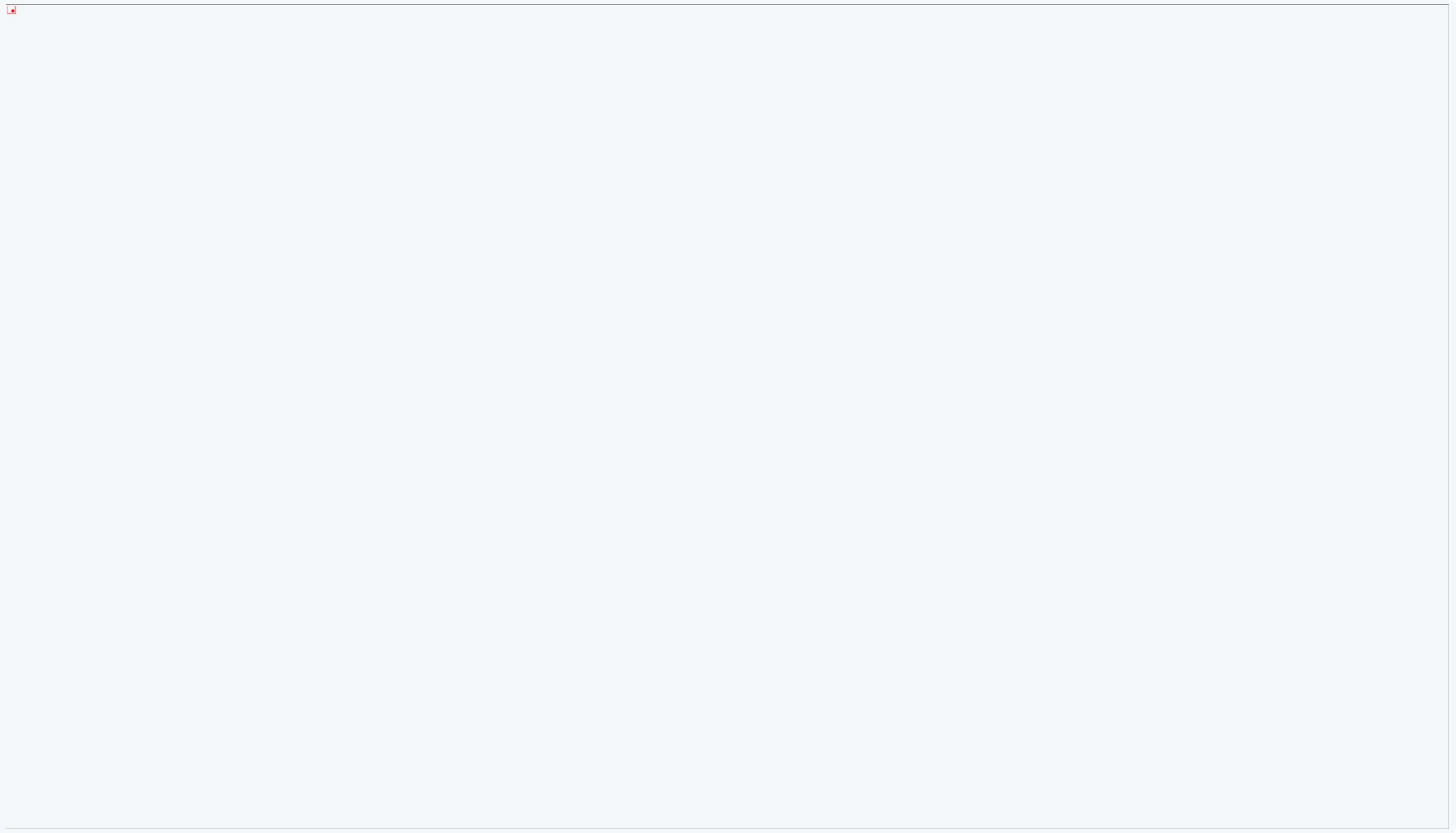
● released ○ no release yet

More information at **5g-mag.com/tech/5g-core/overview**

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# Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



**PROVIDED BY THE REFERENCE TOOLS**

1. **open5gs**
2. **srsRAN_4G**
3. **rt-srsRAN_Project**

Boxes without a badge are not mapped to a repository.

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